HTML Cheat Sheet

Structure: what is on the page. Save (Ctrl+S) and refresh (F5) after every change.

Page skeleton

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Tab title</title>
  <link rel="stylesheet" href="style.css">   <!-- connects your CSS -->
</head>
<body>  ... everything visible goes here ...  </body>
</html>
```

Anatomy of an element

```
<a href="https://github.com" target="_blank">My GitHub</a>
 ^   ^----- attribute -----^                  ^ content   ^ closing tag
 opening tag      (name="value")
```

Everyday tags

Tag	What it does	Example
<h1>...<h6>	Headings (h1 = main title, use once)	`<h1>My Name</h1>`
<p>	Paragraph	`<p>Hello there.</p>`
<a href>	Link	`<a href="page.html">Click</a>`
<img>	Image (no closing tag)	`<img src="me.jpg" alt="Me">`
	Bulleted / numbered list and its items	`<ul><li>HTML</li></ul>`
<button>	Button	`<button>Click Me</button>`
 	Line break	`Line one Line two`
	Important (bold) / emphasis (italic)	`<strong>Warning</strong>`

Grouping tags (semantic)

Tag	Use
`<div>`	Generic box
`<header>`	Top of page/section
`<nav>`	Navigation links
`<main>`	Main content (once)
`<section>`	A themed part
`<footer>`	Bottom of page

Attributes you will use

Attribute	Used for
`href`	Where a link goes
`src`	Image file location
`alt`	Image description
`class`	Name for CSS (reusable)
`id`	Unique name (one element)
`target="_blank"`	Open link in new tab

Golden rules

- Close what you open, **last opened first closed**.
- Indent children by 2 spaces.
- Quotes around attribute values.
- Tag names in lowercase.
- File names: no spaces, exact capitals.

If something is wrong

- Image missing? Check the name, extension and folder.
- Styles missing? Check the `<link>` line and file name.
- Layout broken? Look for a missing closing tag.
- Press **F12** to inspect the page.

CSS Cheat Sheet

Appearance: how it looks. Each rule picks elements and sets their properties.

Anatomy of a rule

```
selector {                  h1 { color: #6941c6; }       /* every h1          */
  property: value;          .card { padding: 20px; }     /* class: dot + name */
}                           #logo { width: 80px; }       /* id: hash + name   */
                            a:hover { color: red; }      /* mouse over it     */
```

Colors, text and fonts

Property	Example
`color`	#1f1637 / red / rgb(0,0,0)
`background`	#f3f0ff
`font-family`	Arial, sans-serif
`font-size`	18px
`font-weight`	bold
`font-style`	italic
`text-align`	left | center | right
`line-height`	1.6
`text-decoration`	none (removes underline)

Size and spacing

Property	Example
`width / height`	300px, 50%
`max-width`	420px
`padding`	20px (inside)
`margin`	16px (outside)
`margin`	0 auto (center box)
`border`	2px solid #6941c6
`border-radius`	12px, 50% = circle
`box-shadow`	0 4px 12px rgba(0,0,0,.15)
`object-fit`	cover (image fills box)

The box model: every element is a box

Layer (inside to outside)	Meaning	Property
Content	The text or image itself	`width, height`
Padding	Space between content and border	`padding`
Border	Line around the padding	`border`
Margin	Space outside, pushing neighbors away	`margin`

Shortcuts and units

```
margin: 10px;                /* all four sides             */
margin: 10px 20px;          /* top+bottom | left+right    */
margin: 5px 10px 15px 20px; /* top right bottom left (clockwise) */
px = pixels (fixed)    % = share of the parent    rem = relative to base text size
```

Layout starter: flexbox

```
.row {
  display: flex;
  gap: 16px;
  justify-content: center;
  align-items: center;
}
```

Hover and a button

```
button {
  background: #6941c6;
  color: white;
  padding: 10px 20px; border: 0;
  border-radius: 8px;
}
button:hover { background: #4c2a99; }
```

Connect HTML and CSS

```
<p class="note">Hi</p>   <!-- HTML, no dot -->
.note { color: gray; }   /* CSS, with dot */
```

Which rule wins?

- A **class** beats a tag.
- If equal, the rule **written later** wins.
- Inspect with F12 to see what applies.

Common mistakes: forgetting `;` or `:` | unmatched `{ }` | writing the dot in HTML or forgetting it in CSS | not saving before refreshing. Learn more at **developer.mozilla.org** (MDN).